

MC224 Parallel and distributed algorithms

- **Course placement:** Parallel and distributed algorithms is a core course for second year students of B.Tech MNC program.
- **Pre-requisite:** Data Structures and Design and Analysis of Algorithms.
- **Course format:** It is 3 hours lecture and 1 hour tutorial every week.
- **Course content:** The aim of the course is introducing the main concepts and methods in parallel and distributed algorithm design, together with performance evaluation and comparison with sequential algorithms. A wide range of topics are discussed in depth, including lists and trees, searching and sorting, graphs, and arithmetic computations.
- **Text book:**
 1. V. Kumar, A Grama, A. Gupta, and G. Karypis, Introduction to Parallel Computing: Design and Analysis of Algorithms.
 2. Joseph Jaja, An Introduction to Parallel Algorithms, Addison Wesley.
 3. J. Suomela, Distributed Algorithms.
- **Assessment method:** Scribe notes, two in-semester examinations and a final examination.
- **Course Outcomes:** Students who complete the course should have the ability to do the following:
 - Able to tell the parallel world from the distributed world.
 - Apply techniques and methods presented along the course aiming to design efficient parallel and distributed algorithms.

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
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- Able to analyze required computational resources, in order to assess performance and correctness of algorithms.

Topics covered:

- **Parallel Algorithms:** Modelling Parallel Computations - Multiprocessor models, Work-depth models, Emulation among models; Parallel Algorithmic Techniques - Divide-and-conquer, Randomization, Parallel pointer techniques; Parallel algorithms for: Prefix sum, searching, sorting, graph problems, Parallel complexity classes: NC and P-completeness.
- **Distributed Algorithms:** Modelling Distributed Computations; Local symmetry-breaking techniques; Distributed algorithms for routing and leader election, Distributed graph algorithms, Lower bounds for distributed problems.